



***SDP 100***

# **Project AgroResQ**

*Intelligent Disaster Relief & Smart Agriculture Ecosystem*

## **Project Proposal**

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| # | Team Member               | Student ID  |
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## 1. Project Title

### Project AgroResQ: Intelligent Disaster Relief & Smart Agriculture Ecosystem

A unified software decision-support platform that combines intelligent disaster relief management with data-driven agricultural recovery, designed specifically for disaster-prone regions of Bangladesh.

## 2. Team Members

| # | Name                      | Student ID  | Role                          |
|---|---------------------------|-------------|-------------------------------|
| 1 | S. M. Masrur Rafin        | 20254103164 | Disaster Module Lead          |
| 2 | Abu Hourira MD Abdul Aziz | 20254103163 | Victim & Shelter Management   |
| 3 | MD Tanvir Ahmmed          | 20254103191 | Agriculture Module Lead       |
| 4 | Ishraq Ahmmed             | 20254103173 | Agriculture Data & Algorithms |
| 5 | Suborna Akter             | 20254103352 | Testing & Documentation       |

## 3. Problem Statement

Bangladesh is one of the most disaster-prone countries in the world, regularly affected by floods, cyclones, river erosion, landslides, and fires. During such emergencies, local authorities and relief organizations struggle to identify vulnerable victims, prioritize rescue operations, allocate shelters fairly, distribute relief materials efficiently, and coordinate volunteers across affected areas. Because most of this coordination is still done manually using paper records and phone calls, response times are slow, resources are often misallocated, and the most vulnerable victims do not always receive timely help.

The damage does not end once the immediate emergency passes. Disasters routinely destroy standing crops, degrade soil quality through salinity and waterlogging, and disrupt irrigation infrastructure, placing long-term food security at risk. Farmers in affected regions are frequently left without reliable guidance on which crops are viable to replant, how to restore soil health, or how to manage irrigation under changed field conditions, which slows agricultural and economic recovery for months after the disaster itself has ended.

There is therefore a clear need for a single intelligent system that addresses both halves of this problem: supporting emergency responders during the disaster itself, and supporting farmers as they rebuild their land and livelihoods afterward. Local government disaster offices, NGOs, relief volunteers, and smallholder farmers in flood- and cyclone-affected districts are the primary beneficiaries of such a system.

## 4. Project Objectives

The primary goals and expected outcomes of this project are:

1. To efficiently manage disaster information and affected victims through a centralized digital system.
2. To prioritize rescue operations automatically based on calculated risk and vulnerability levels.
3. To allocate shelters and relief resources intelligently, minimizing waste and delay.

4. To coordinate rescue teams and volunteers based on availability, skill, and proximity.
5. To monitor and record soil conditions through a software-based data entry and tracking system.
6. To recommend suitable crops for replanting based on post-disaster soil and weather conditions.
7. To provide irrigation planning and soil analysis support for affected farmland.
8. To predict crop diseases and agricultural risks before they spread.
9. To generate consolidated reports and dashboards that support data-driven decision-making.
10. To improve overall community resilience and long-term food security in disaster-prone areas.

## 5. Expected Features

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### 5.1 Disaster Management Features

- Disaster Registration - log disaster type, location, severity, and affected area.
- Victim Registration - capture individual and family details of affected persons.
- Family Information Management - link victims into household/family units.
- Vulnerability Assessment - score victims by age, health, disability, and exposure risk.
- Rescue Priority Scheduling - queue rescue requests using a priority algorithm.
- Shelter Allocation - assign victims to nearest available shelter based on capacity.
- Resource Allocation - track and distribute food, water, medicine, and supplies.
- Relief Distribution - log and verify delivery of relief materials to victims.
- Volunteer Coordination - match volunteers to tasks based on skills and location.
- Missing Person Tracking - register and search for missing individuals.
- Emergency Alert System - broadcast warnings and updates to registered users.
- Route Optimization - compute efficient rescue and delivery routes using graph algorithms.
- Budget Estimation - estimate relief operation costs from resource and victim data.
- Analytics Dashboard - visualize disaster impact, response status, and resource gaps.
- Intelligent Recommendations - suggest response actions based on historical patterns.

### 5.2 Agriculture Features

- Crop Recommendation - suggest viable crops based on soil and seasonal data.
- Soil Analysis - evaluate soil moisture, pH, and nutrient indicators.
- Irrigation Planning - generate watering schedules from soil moisture readings.
- Crop Disease Prediction - flag likely disease risk from environmental data.
- Soil Data Tracking - record and update soil moisture, pH, and nutrient values manually or via imported data.
- Post-Disaster Agricultural Recovery Support - guided recovery steps for damaged farmland.

## 6. Workflow Diagram

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The diagram below illustrates the overall workflow of AgroResQ, showing how inputs from the field (disaster reports and soil data entries) flow through processing modules to produce actionable outputs for both emergency responders and farmers.

### INPUTS

Disaster Reports • Victim Data • Soil Data Entries • Weather Data



### PROCESSES

Disaster Module: Risk Assessment & Priority Scheduling



Resource Module: Shelter & Relief Allocation, Route Optimization



Agriculture Module: Soil Analysis, Crop & Irrigation Recommendation



Decision Support Engine: Combines disaster + agriculture data into recommendations



### OUTPUTS

Rescue & Relief Action Plans • Crop & Irrigation Plans • Reports & Dashboards

## 7. Flowchart

The flowchart below represents the logical sequence of operations within the system, from disaster onset through to final reporting.

Start



Disaster Occurs



Collect Disaster Data



Register Victims & Families





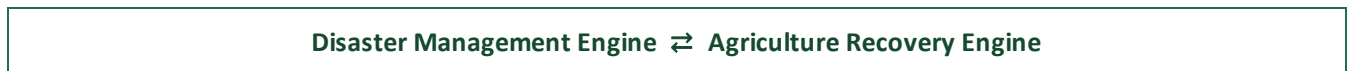
## 8. System Architecture

AgroResQ follows a layered architecture that separates data collection, processing logic, and presentation, allowing the disaster and agriculture modules to operate independently while sharing a common data and reporting layer.

### Layer 1 - Data Input



### Layer 2 - Core Application Logic (C/C++)



### Layer 3 - Algorithms & Data Structures

Priority Queue • Scheduling Algorithm • Graph-based Route Optimization • Decision Support Logic



#### Layer 4 - Data Storage

File Handling



#### Layer 5 - Output & Reporting

Reports • Analytics Dashboard • Recommendations • Alerts

The Disaster Management Engine and Agriculture Recovery Engine communicate through shared data structures so that, for example, a region flagged as flood-affected automatically becomes a candidate for post-disaster soil analysis and crop recommendation once the immediate rescue phase is complete.

## 9. Required Technologies

### 9.1 Programming Language

- C / C++

### 9.2 Database

- File Handling

### 9.3 Tools

- Visual Studio Code
- Code::Blocks
- Git & GitHub

### 9.4 Concepts & Algorithms

- Data Structures (Queues, Linked Lists, Graphs)
- Priority Queue
- Scheduling Algorithm
- Graph Algorithm for Route Optimization
- Decision Support System

## 10. Task Distribution Among Team Members

| Member                    | Responsibility              |
|---------------------------|-----------------------------|
| S. M. Masrur Rafin        | Disaster Module Development |
| Abu Hourira MD Abdul Aziz | Victim & Shelter Management |

| Member           | Responsibility                                               |
|------------------|--------------------------------------------------------------|
| MD Tanvir Ahmmed | Agriculture Module                                           |
| Ishraq Ahmmed    | Agriculture Data & Algorithms (Crop, Soil, Irrigation Logic) |
| Suborna Akter    | Testing, Documentation & Report Generation                   |

## 11. Project Timeline

| Week   | Tasks                                  |
|--------|----------------------------------------|
| Week 1 | Requirement Analysis & Design          |
| Week 2 | Database and Class Design              |
| Week 3 | Disaster Management Module             |
| Week 4 | Agriculture Module                     |
| Week 5 | Crop & Irrigation Recommendation Logic |
| Week 6 | Algorithm Implementation               |
| Week 7 | Testing & Bug Fixing                   |
| Week 8 | Documentation & Final Presentation     |

## 12. Sustainable Development Goals (SDGs)

- SDG 2: Zero Hunger - through crop recommendation and post-disaster agricultural recovery.
- SDG 3: Good Health and Well-being - through faster, better-coordinated disaster response.
- SDG 11: Sustainable Cities and Communities - through organized shelter and resource management.
- SDG 13: Climate Action - through improved resilience to climate-related disasters.

## 13. Expected Outcome

AgroResQ will deliver an integrated software platform that strengthens disaster response while accelerating agricultural recovery in its aftermath. Emergency authorities will be able to register and prioritize victims, allocate shelters and resources, and coordinate volunteers far more efficiently than manual processes allow. Once the immediate crisis subsides, farmers in the same affected region will gain access to soil analysis, crop recommendations, and irrigation guidance generated from recorded field data, helping restore productivity faster. Together, these capabilities are expected to reduce response delays, improve resource utilization, and strengthen long-term food security and community resilience in disaster-prone parts of Bangladesh.